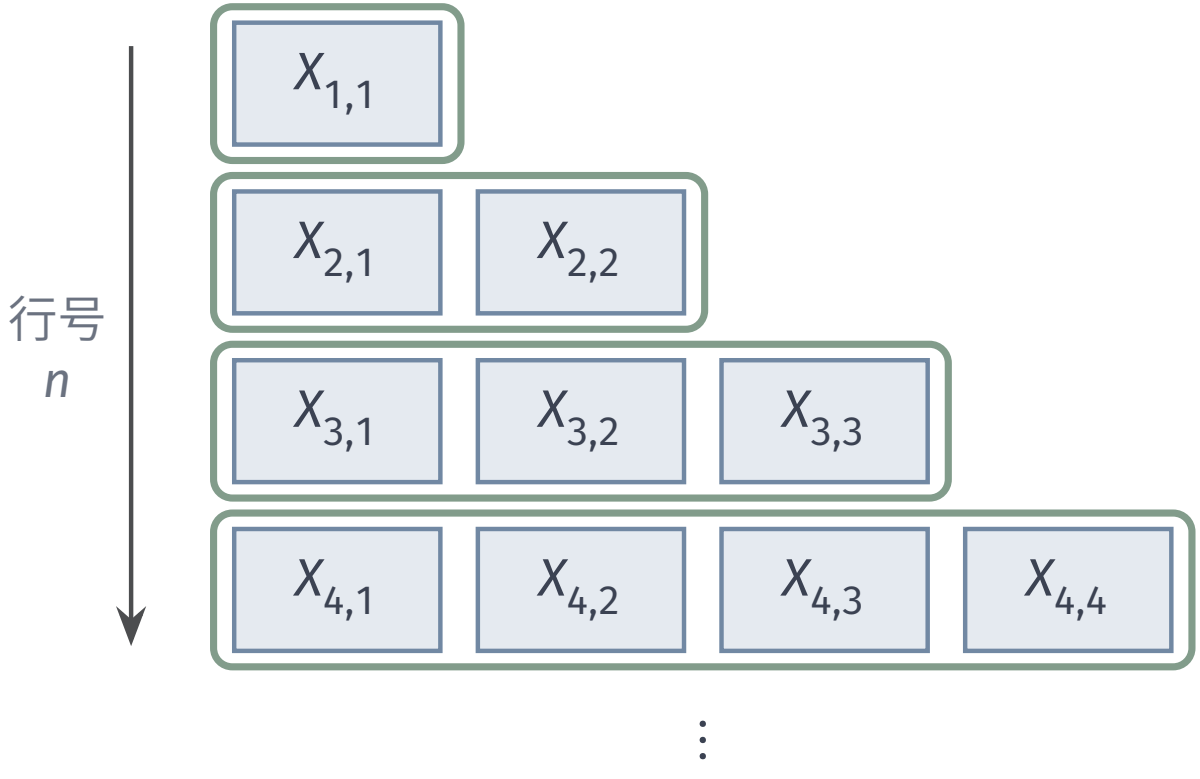

$$S_1 = \sum_k X_{1,k},\ s_1^2 = \mathrm{Var}(S_1)$$

$$S_2 = \sum_k X_{2,k},\ s_2^2 = \mathrm{Var}(S_2)$$

$$S_3 = \sum_k X_{3,k},\ s_3^2 = \mathrm{Var}(S_3)$$

$$S_4 = \sum_k X_{4,k},\ s_4^2 = \mathrm{Var}(S_4)$$

同一行：相互独立

不同行：不作要求